

Quick Check

Lesson 4: Hormones as the Control Switches of Moulting and Metamorphosis

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 4

A scientist removes the brain neurosecretory cells (the cells that make Brain Hormone) from a young caterpillar. What do you predict will happen to the caterpillar's moulting?

- A** It will stop moulting because the signal that starts the process is missing
- B** It will turn into a pupa straight away
- C** It will moult more often because the brain no longer slows growth
- D** It will moult as normal because the shell splits by itself

Which gland releases Ecdysone after it receives the signal from Brain Hormone?

A Prothoracic glands

B Corpora allata

C Malpighian tubules

D Salivary glands

In an experiment, a larva has its brain neurosecretory cells removed and it stops moulting. The same larva is then injected with Ecdysone and it starts to moult again. What does this result show?

- A** Ecdysone is made by the brain neurosecretory cells
- B** Ecdysone triggers moulting and acts after the brain signal
- C** Moulting does not need any hormones
- D** Ecdysone keeps the larva in the larval form

In an experiment, the corpora allata (which make Juvenile Hormone) are removed from a young larva. It changes early into a small pupa. What does this show about Juvenile Hormone?

- A** Juvenile Hormone is made by the prothoracic glands
- B** Juvenile Hormone is needed to trigger every moult
- C** Juvenile Hormone keeps the larval state, so low Juvenile Hormone allows pupation
- D** High Juvenile Hormone makes a larva change into a pupa early

A farmer in the Rift Valley has an armyworm larva that is about to make its last larval moult. Suppose Juvenile Hormone stayed high at this moult. What would you expect to happen?

- A** It would moult straight into an adult moth
- B** It would stop moulting because Ecdysone is blocked completely
- C** It would moult into a pupa as normal
- D** It would moult into another larva instead of a pupa

A student annotates a stepped growth curve for a grasshopper. Each sudden upward step in body size marks a moult. What should the student write about Ecdysone at each step?

- A** Ecdysone peaks at the step, because it triggers moulting
- B** Ecdysone is lowest at the step, because it keeps the old cuticle in place
- C** Ecdysone is only made after the insect becomes an adult
- D** Ecdysone is absent during the whole life cycle

Which sequence correctly shows how hormones control a moult in an insect?

- A** Corpora allata, then Ecdysone, then Brain Hormone, then moult
- B** Brain Hormone, then the prothoracic gland releases Ecdysone, then moult; Juvenile Hormone decides the type of moult
- C** Ecdysone, then the prothoracic gland, then Brain Hormone, then moult
- D** Juvenile Hormone, then Brain Hormone, then the prothoracic gland, then Ecdysone

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).